Shivam Kumar Verma

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Experience

poiro.com **Jul 2026 – Present**
Fullstack Developer Intern *Python, FastAPI, MongoDB, Redis, Celery, Next.js, TypeScript, Cloudflare R2, Turborepo*

- **DAG Workflow Engine:** Built the node and group system behind the AI workflow builder: each workflow is a DAG of generation steps passing typed outputs to the next, and groups bundle steps into reusable sub-workflows that run as one unit.
- **Digital Asset Service:** Designed the data model and APIs for one shared asset service that replaced separate storage in **3 apps**, and migrated all existing assets into it; added folders, reusable prompt Elements, and upload-time duplicate blocking with pHash and dHash image similarity.
- **Provider Adapter Layer:** Integrated Pixabay, Unsplash, and Pinterest behind one common adapter interface, with per-org Redis caching to avoid repeat API calls and spend caps to stop runaway provider costs.

klimb.io **Nov 2025 – May 2026**
Full Stack Developer Intern *TypeScript, Express.js, PostgreSQL, Redis, AWS*

- **Usage-Based Billing Ledger:** Replaced on/off feature flags with a credit ledger that meters each customer's usage of **6+ AI features**, which made usage-based pricing possible.

Projects

Yuvro | GitHub | Demo | *Replit (Web IDE), Python, FastAPI, PostgreSQL, Kubernetes, AWS S3, WebSockets*

- **Content-Addressed Storage:** Stores each file once by its content hash, so identical files across projects are never duplicated, with a scheduled garbage collector that deletes files no project references.
- **Per-Project Containers on Kubernetes:** Starts an isolated container for each project on demand, with a live WebSocket terminal, file editor, app preview, and database browser, so user code never runs on the main servers.

CodeSM | GitHub | Live | *Code Platform, Node.js, Express.js, BullMQ, Redis, PostgreSQL/Drizzle, Docker, AWS S3*

- **Sandboxed Code Execution:** Runs every submission in a locked-down container with no network, a read-only filesystem, and CPU, memory, and time limits, so infinite loops, fork bombs, and network calls stay contained.
- **Async Grading Queue:** Moved grading onto BullMQ workers so the API responds instantly; each submission reuses one container across its test cases and stops at the first failed test.

DeepDoc | GitHub | *RAG, Next.js, TypeScript, Azure OpenAI, Pinecone, PostgreSQL/Drizzle, Vercel Blob*

- **RAG Pipeline:** Splits each PDF into section-aware chunks by topic, searches only that document's vectors, and rewrites follow-up questions using chat history so they retrieve the right sections.
- **Retrieval Ranking and Tuning:** Ranks matches by score, drops weak and duplicate chunks, and packs the best into a token budget; tuned these settings across **100+ configurations** and fixed a bug that returned empty results for every query.

Skills

Languages: C++, Python, TypeScript, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Celery, BullMQ
Data & Infrastructure: PostgreSQL, MongoDB, Redis, Pinecone, Docker, Kubernetes, AWS (EC2, S3)
AI Engineering: RAG, embeddings and vector search, query rewriting, LLM evaluation
Frontend & Tooling: React, Next.js, Git, Linux, CI pipelines

Education

Indian Institute of Information Technology Nagpur **Nov 2022 – May 2026**
B.Tech in Computer Science and Engineering *Nagpur, Maharashtra*

Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef (1700+) · Pupil on Codeforces.